Technical Report Writing

on

Fundamentals of TREE

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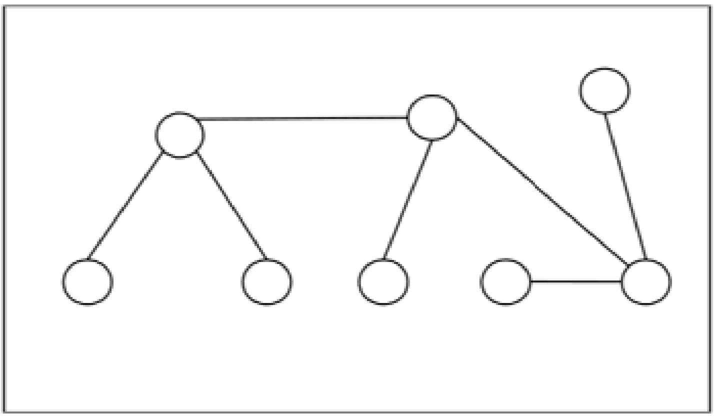
Abstract:

Trees are fundamental structures in discrete mathematics and graph theory, offering a versatile framework for representing hierarchical relationships and organizing data efficiently. In this technical report, we explore the foundational concepts of trees, their properties, and their applications in various domains.

Introduction:

Definition − A Tree is a connected acyclic undirected graph. There is a unique path between every pair of vertices in G. A tree with N number of vertices contains (N−1) number of edges. The vertex which is of 0 degree is called root of the tree. The vertex which is of 1 degree is called leaf node of the tree and the degree of an internal node is at least 2.

Example − The following is an example of a tree −



A connected graph without any circuit is called a Tree. In other words, a tree is an undirected graph G that satisfies any of the following equivalent conditions:

- Any two vertices in G can be connected by a unique simple path.
- G is acyclic, and a simple cycle is formed if any edge is added to G.
- G is connected and has no cycles.
- G is connected but would become disconnected if any single edge is removed from G.

Procedure and Discussion:

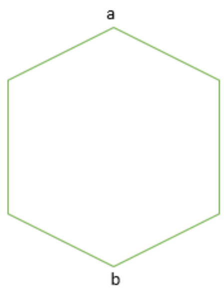
A tree or general trees is defined as a non-empty finite set of elements called vertices or nodes having the property that each node can have minimum degree 1 and maximum degree n. It can be partitioned into n+1 disjoint subsets such that the first subset contains the root of the tree and remaining n subsets includes the elements of the n subtree.

Theorems Of Tree

Some theorems related to trees are:

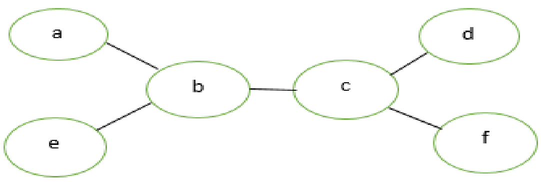
Theorem 1: Prove that for a tree (T), there is one and only one path between every pair of vertices in a tree.

Proof: Since tree (T) is a connected graph, there exist at least one path between every pair of vertices in a tree (T). Now, suppose between two vertices a and b of the tree (T) there exist two paths. The union of these two paths will contain a circuit and tree (T) cannot be a tree. Hence the above statement is proved.



Theorem 2: If in a graph G there is one and only one path between every pair of vertices than graph G is a tree.

Proof: There is the existence of a path between every pair of vertices so we assume that graph G is connected. A circuit in a graph implies that there is at least one pair of vertices a and b, such that there are two distinct paths between a and b. Since G has one and only one path between every pair of vertices. G cannot have any circuit. Hence graph G is a tree



Theorem 3: Prove that a tree with n vertices has (n-1) edges.

Proof: Let n be the number of vertices in a tree (T). If n=1, then the number of edges=0. If n=2 then the number of edges=1. If n=3 then the number of edges=2. Hence, the statement (or result) is true for n=1, 2, 3. Let the statement be true for n=m. Now we want to prove that it is true for n=m+1. Let be the edge connecting vertices say Vi and Vj. Since G is a tree, then there exists only one path between vertices Vi and Vj. Hence if we delete edge e it will be disconnected graph into two components G1 and G2 say. These components have less than m+1 vertices and there is no circuit and hence each component G1 and G2 have m1 and m2 vertices.

Now, the total no. of edges = (m1-1) + (m2-1) +1

= (m1+m2)-1

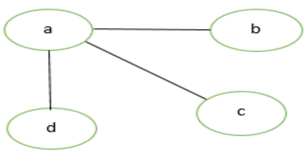
= m+1-1

= m.

Hence for **n=m+1** vertices there are m edges in a tree (T). By the mathematical induction the graph exactly has n-1 edges.

Theorem 4: Prove that any connected graph G with n vertices and (n-1) edges is a tree.

Proof: We know that the minimum number of edges required to make a graph of n vertices connected is (n-1) edges. We can observe that removal of one edge from the graph G will make it disconnected. Thus, a connected graph of n vertices and (n-1) edges cannot have a circuit. Hence a graph G is a tree.



Theorem 5: Prove that a graph with n vertices, **(n-1)** edges and no circuit is a connected graph.

Proof: Let the graph G is disconnected then there exist at least two components G1 and G2 say. Each of the component is circuit-less as G is circuit-less. Now to make a graph G connected we need to add one edge e between the vertices Vi and Vj, where Vi is the vertex of G1 and Vj is the vertex of component G2. Now the number of edges in **G = (n – 1) +1 =n.**

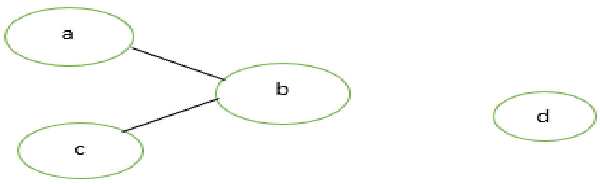


Figure 7: Disconnected Graph

Now, G is connected graph and circuit-less with n vertices and n edges, which is impossible because the connected circuit-less graph is a tree and tree with n vertices has **(n-1)** edges. So the graph G with n vertices, **(n-1)** edges and without circuit is connected. Hence the given statement is proved.

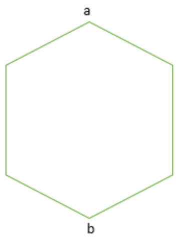
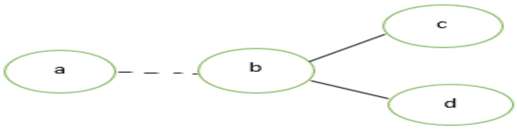


Figure 8: Connected graph G

Theorem 6: A graph G is a tree if and only if it is minimally connected.

Proof: Let the graph G is minimally connected, i.e.; removal of one edge make it disconnected. Therefore, there is no circuit. Hence graph G is a tree. Conversely, let the graph G is a tree i.e.; there exists one and only one path between every pair of vertices and we know that removal of one edge from the path makes the graph disconnected. Hence graph G is minimally connected.



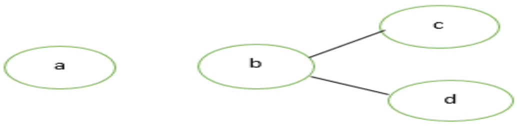


Figure 9: Minimally Connected Graph

Theorem 7: Every tree with at-least two vertices has at-least two pendant vertices.

Proof: Let the number of vertices in a given tree T is n and n>=2. Therefore, the number of edges in a tree T=n-1 using above theorems.

summation of (deg(Vi)) = 2*e

= 2*(n-1)

=2n-2

The degree sum is to be divided among n vertices. Since a tree T is a connected graph, it cannot have a vertex of degree zero. Each vertex contributes at-least one to the above sum. Thus, there must be at least two vertices of degree 1. Hence every tree with at-least two vertices have at-least two pendant vertices.

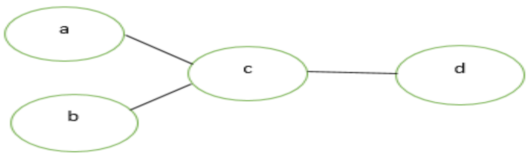


Figure 10: Here a, b and d are pendent vertices of the given graph

Theorem 8: Show that every tree has either one or two centres.

Proof: We will use one observation that the maximum distance max d(v, w) from a given vertex v to any other vertex w occurs only when w is pendant vertex. Now, let T is a tree with n vertices (n>=2) T must have at least two pendant vertices. Delete all pendant vertices from T, then resulting graph T' is still a tree. Again, delete pendant vertices from T' so that resulting T" is still a tree with same centres. Note that all vertices that T had as centres will still remain centres in T'–>T"–>T"'–>…. continue this process until the remaining tree has either one vertex or one edge. So, in the end, if one vertex is there this implies tree T has one centre. If one edge is there then tree T has two centres.

Theorem 9: Prove the maximum number of vertices at level 'L' in a binary tree is 2^L where L>=0.

Proof: The given theorem is proved with the help of mathematical induction. At level 0 (L=0), there is only one vertex at level (L=1), there is only 2^1vertices. Now we assume that statement is true for the level (L-1). Therefore, maximum number of vertices on the level (L-1) is 2^{L-1}. Since we know that each vertex in a binary tree has the maximum of 2 vertices in next level, therefore the number of vertices on the level L is twice that of the level L-1.

Hence at level L, the number of vertices is: - $2^{L-1} * 2^1 = 2^L$.

Applications of Tree:

Here are some applications of trees in discrete mathematics specifically:

1. **Minimum Spanning Trees** (MST): In graph theory, finding a minimum spanning tree of a weighted graph is a fundamental problem. Algorithms like Prim's and Kruskal's algorithms utilize trees to efficiently find the minimum spanning tree, which has applications in network design and optimization problems.
2. Tree Isomorphism: The problem of determining whether two trees are isomorphic is relevant in various areas such as pattern recognition and molecular biology. Algorithms exist to check tree isomorphism efficiently.
3. Graph Colouring and Chromatic Number: Trees are trivially 2-colorable (using only two colours), as they contain no cycles. The chromatic number of a tree is 2. This property is relevant in graph colourings problems, scheduling, and register allocation in compilers.
4. Generating Trees for Algorithms: Trees are used in algorithm design, particularly in backtracking algorithms. For example, generating all possible permutations or combinations of a set can be efficiently done using a tree structure.
5. **Hamiltonian Paths** and Trees: The problem of finding Hamiltonian paths or cycles in trees has applications in optimization and routing problems, such as in the design of computer networks and transportation systems.

6. **Spanning Trees** in Graph Theory: The concept of spanning trees is crucial in graph theory, where trees are used to extract essential information from graphs. Properties of spanning trees and their algorithms are used in network design, routing, and optimization.

These applications demonstrate how trees are not only foundational in discrete mathematics but also play a significant role in solving practical problems across various domains.

Conclusion:

The fundamentals of trees in discrete mathematics form a cornerstone of graph theory and data structure analysis. Trees are hierarchical structures composed of nodes connected by edges, with various types and properties defining their structure and behaviour. Understanding the basics of trees, including binary trees, spanning tree and their specialized variants, is essential for algorithm design, data organization, and problem-solving in computer science and related fields.

Thus, the fundamentals of trees in discrete mathematics encompass a wide array of concepts, properties, and applications. Understanding the fundamentals of trees is essential for tackling combinatorial problems, graph analysis, and algorithmic optimizations in discrete mathematics contexts.

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